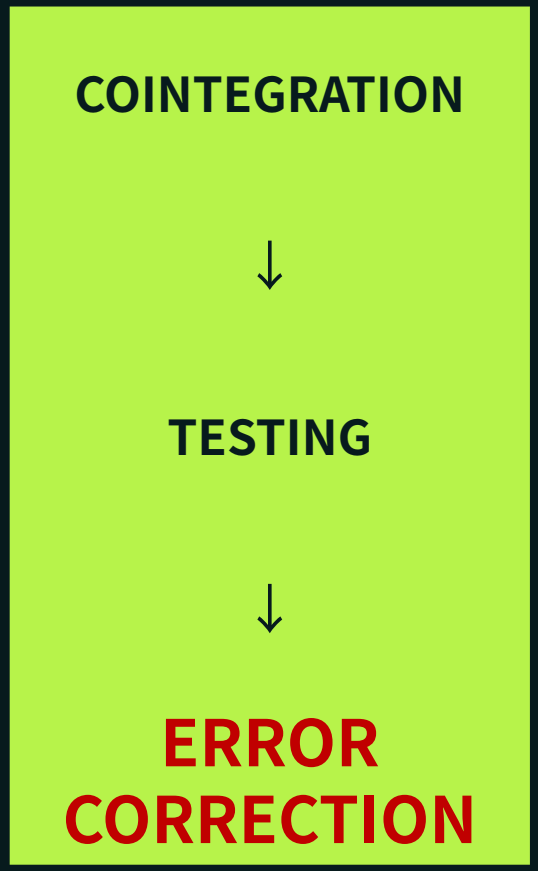
Cointegration & ECM Formula Cheat Sheet

One formula per slide — cointegration · tests · error correction



Advanced Financial Time Series · Week 7

Three parts, formula by formula, in one hour

Each part runs definition → tests and models → pictures, one formula per slide

01	**Cointegration**	**A stationary mix of two wandering series** I(1) · definition · common trend · spurious
02	**Testing for cointegration**	**Prove it with a test, not a chart** Engle-Granger · critical values · Johansen
03	**Error correction and trading**	**Measure the pull back, then trade it** ECM · VECM · half-life · z-score · breakdown
04	**Synthesis**	**Concept → test → trade** Tests compared · pairs-trading pipeline

Find the anchor with cointegration → prove it with a test → trade it with error correction

01

Cointegration

Two series that each wander, but one combination stays put

A random walk is I(1); its difference is I(0)

Order of integration: how many differences make it stationary — the basic idea behind cointegration

Random walk and its difference

$$y_t = y_{t-1} + \varepsilon_t \;\Rightarrow\; y_t \sim I(1)$$
$$\Delta y_t = y_t - y_{t-1} = \varepsilon_t \sim I(0)$$

How to read it

- I(1): shocks pile up, no mean to return to — price levels
- I(0): moves around a fixed mean — returns
- Differencing makes it stationary but throws away the levels
- Week 6's VAR used differenced series

Cointegration recovers the long-run link in levels that differencing threw away

Cointegrated: a stationary mix of two I(1) series

Definition by Engle and Granger (1987) — Nobel Prize in Economics, 2003

Definition

$$X_t \sim I(1), \quad Y_t \sim I(1), \qquad z_t = Y_t - \beta X_t \sim I(0)$$

Cointegrating vector (1, −β)

$$z_t = \underbrace{\begin{pmatrix} 1 & -\beta \end{pmatrix}}_{\text{cointegrating vector}} \begin{pmatrix} Y_t \\ X_t \end{pmatrix} = Y_t - \beta X_t$$

How to read it

- X and Y each wander, but the spread z is stationary
- It holds for one particular β only
- z is the equilibrium error — the spread we watch
- In trading, β becomes the hedge ratio

The heart of cointegration analysis is finding the β that makes the spread stationary

A stationary mix needs one shared trend

Mixing two I(1) series usually gives I(1) again — unless they share the same stochastic trend

Usually impossible

$$X_t \sim I(1),\ Y_t \sim I(1) \implies aX_t + bY_t \sim I(1) \quad \text{(in general)}$$

Common-trend form

$$\tau_t = \tau_{t-1} + \eta_t \qquad \text{(common stochastic trend)}$$
$$X_t = \tau_t + u_t, \qquad Y_t = \beta\,\tau_t + v_t$$
$$z_t = Y_t - \beta X_t = v_t - \beta\,u_t \ \sim\ I(0)$$

How to read it

- τ: the random walk both series share
- u, v: their own stationary noise
- In Y − βX, the βτ cancels exactly
- Cointegration = fewer trends than series

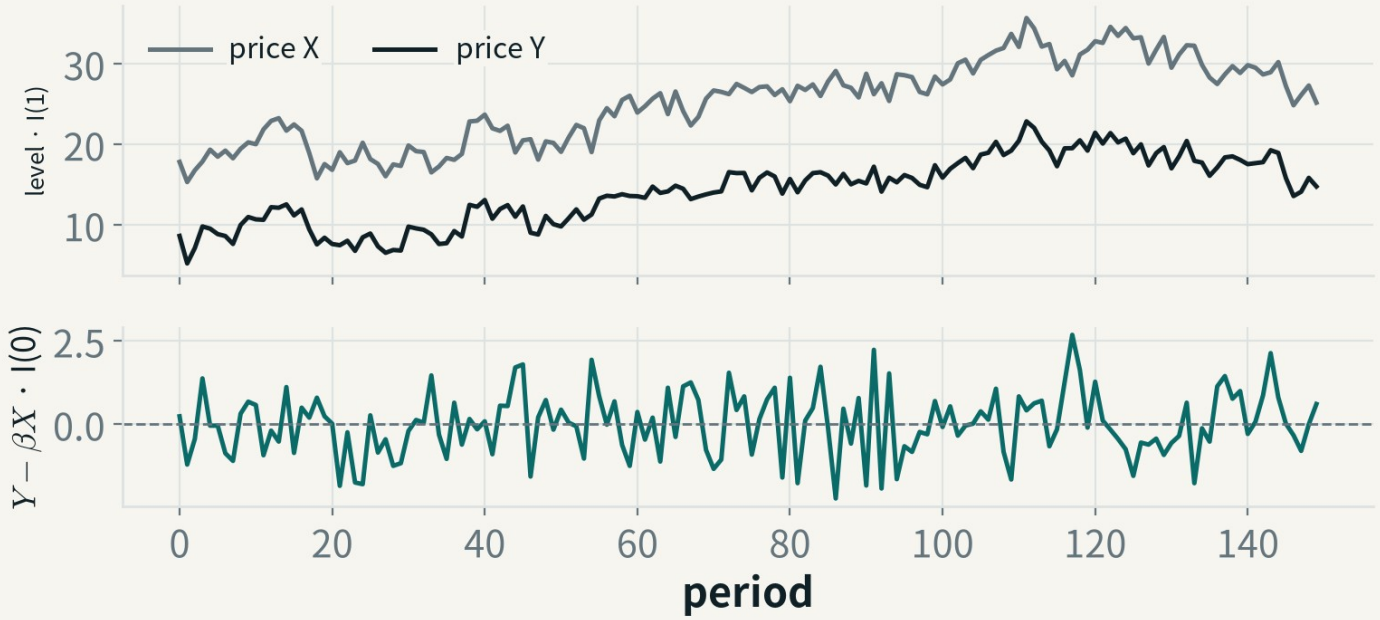
If each has its own trend, no combination can remove them

Prices wander, but the spread returns to 0

A drunk and her dog: each a random walk, but the leash ties their distance (simulation, illustrative data)

THE LEASH

- Top: price levels, each I(1)
- Bottom: the spread, I(0)
- Estimated $\beta \approx 0.79$
- Tied together, but not fixed



A gap that returns to its mean is the long-run equilibrium

Stationary residuals mean the regression is not spurious

The exact exception to Week 2's warning 'do not regress trends on trends'

The residuals decide

$$Y_t = \alpha + \beta X_t + e_t : \quad \begin{cases} e_t \sim I(1) & \Rightarrow \text{ spurious} \\ e_t \sim I(0) & \Rightarrow \text{ cointegrated} \end{cases}$$

How to read it

- Spurious: high R^2, but the residuals wander
- Cointegrated: stationary residuals — proof of a real link
- So testing the residuals for stationarity tells them apart
- That test is Part 2's Engle-Granger

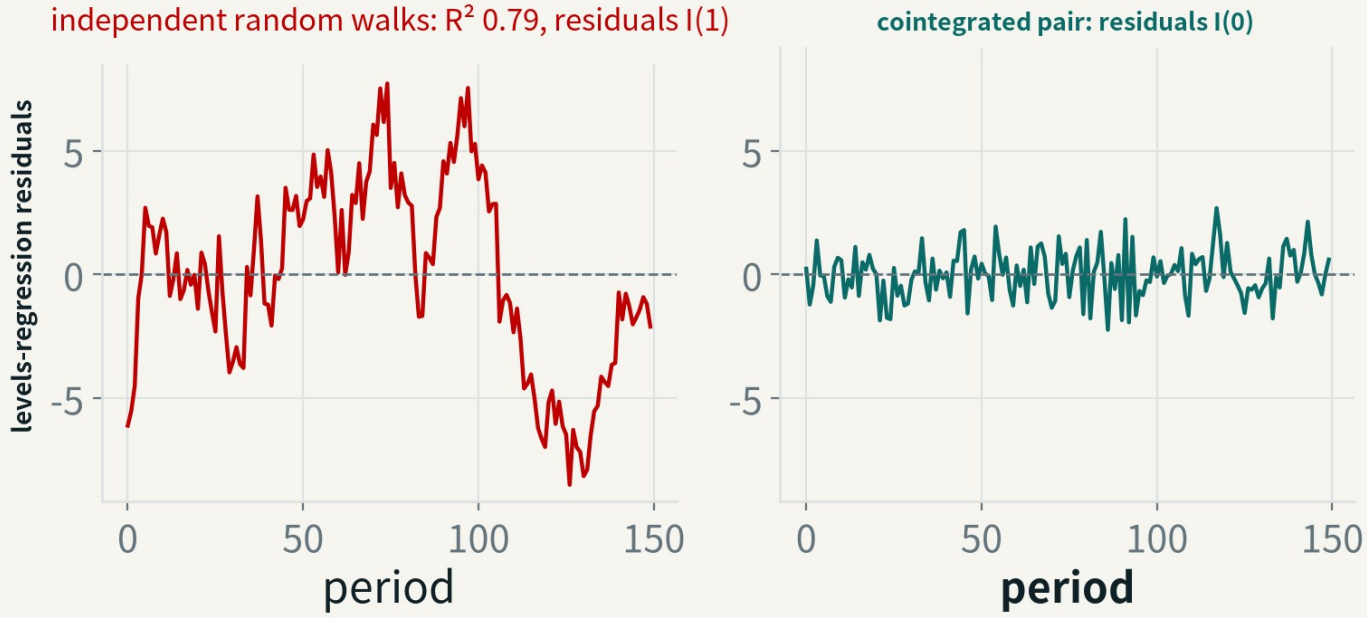
Stationary residuals certify a real relationship

Spurious residuals wander; cointegrated ones return

Same vertical axis, 150 periods of levels-regression residuals (simulation, illustrative data)

SPURIOUS vs GENUINE

- Even two independent random walks get $R^2 = 0.79$
- But the residuals drift far to one side
- Cointegrated residuals stay in a narrow band around 0



Look at the residuals, not R^2

Not correlation — and the spread is a trading signal

Correlation: returns move together short term · cointegration: price levels share a long-run equilibrium

Correlation vs cointegration

$$\rho = \mathrm{Corr}(\Delta X_t,\, \Delta Y_t) \quad \text{vs.} \quad Y_t - \beta X_t \sim I(0)$$

The spread as a trade

$$z_t > \mu_z : \ \text{short } Y,\ \text{long } \beta X$$
$$z_t < \mu_z : \ \text{long } Y,\ \text{short } \beta X$$

How to read it

- High correlation does not imply cointegration
- Candidates: spot–futures, dual listings, two firms with the same driver
- No economic force tying them, no cointegration
- For a non-pair, the 'spread' is a random walk

Market-neutral — buy one, sell the other

02

Testing for cointegration

With a test, not a chart — Engle-Granger two steps and the Johansen system test

Check each series is I(1), then fit the long run

Step 0: unit-root tests → Engle-Granger step 1: OLS

Step 0: check the order of integration

$$X_t \sim I(1) \iff \begin{cases} \text{ADF on } X_t: & \text{not reject} \\ \text{ADF on } \Delta X_t: & \text{reject} \end{cases}$$

Step 1: levels regression and residuals

$$\text{Step 1:} \quad Y_t = \alpha + \beta X_t + u_t \xrightarrow{\text{OLS}} \hat{u}_t = Y_t - \hat{\alpha} - \hat{\beta} X_t$$

How to read it

- ADF null: a unit root (non-stationary)
- If already I(0) or I(2), cointegration is the wrong tool
- Slope $\hat{\beta}$ = cointegrating coefficient = hedge ratio
- Residual $\hat{u}$ = estimated equilibrium error (the spread)

Getting the order of integration right is step 0 of any cointegration test

Step 2 asks whether the residuals have a unit root

The null is 'not cointegrated' — reject it to claim cointegration

Step 2: ADF regression on the residuals

$$\Delta \hat{u}_t = \gamma\, \hat{u}_{t-1} + \sum_{i=1}^{p} \phi_i\, \Delta \hat{u}_{t-i} + e_t, \qquad H_0 : \gamma = 0$$

Decision

$$t_\gamma = \frac{\hat{\gamma}}{\mathrm{se}(\hat{\gamma})} \;<\; c_{\mathrm{EG}} \;\Rightarrow\; \text{reject } H_0 \;\Rightarrow\; \text{cointegrated}$$

How to read it

- γ = 0: residuals are a random walk — spurious
- γ < 0: deviations are pulled back to 0 — cointegrated
- The ϕ terms absorb residual autocorrelation
- Pick the lag p by an information criterion

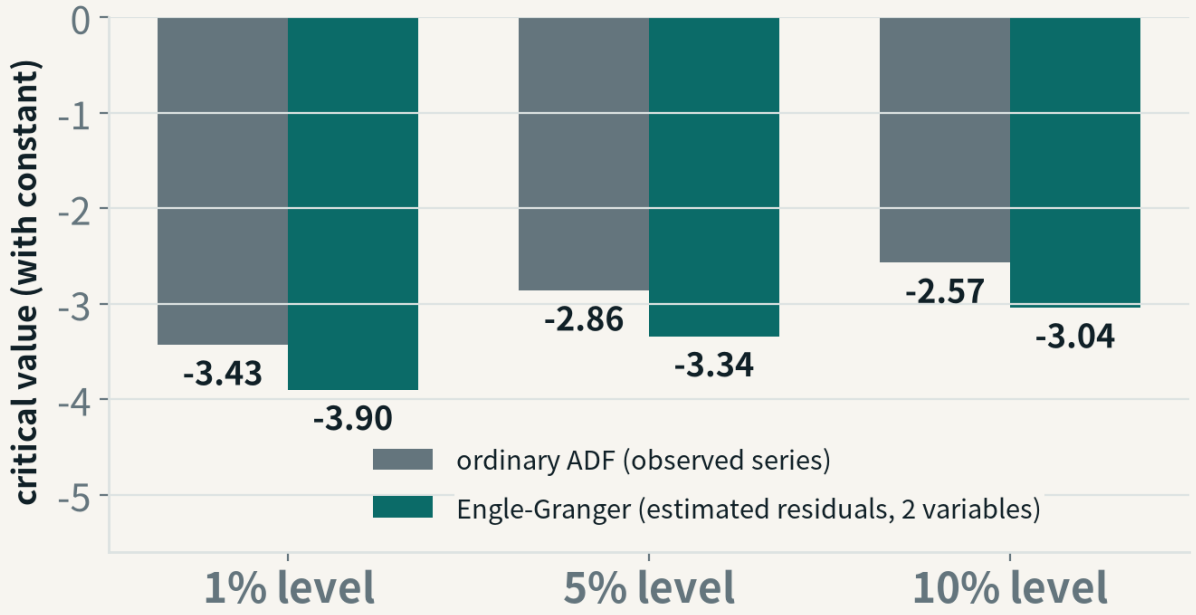
Everything rests on this one stationarity test

Estimated residuals need stricter critical values

MacKinnon asymptotic critical values, with constant — more negative with more variables

CRITICAL VALUES

- OLS picks β to make the residuals small
- So the residuals look more stationary than they are
- Ordinary ADF values find false cointegration
- Check both orders, Y~X and X~Y



Never use ordinary ADF critical values in step 2

Johansen counts equilibria by the rank of Π

A VAR rewritten in differences, the VECM — every variable treated alike

VECM form

$$\Delta \mathbf{Y}_t = \Pi\, \mathbf{Y}_{t-1} + \sum_{i=1}^{p-1} \Gamma_i\, \Delta \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t, \quad r = \operatorname{rank}(\Pi)$$

With rank r, a product of two matrices

$$\Pi = \underbrace{\alpha}_{k \times r}\ \underbrace{\beta^{\top}}_{r \times k}$$

How to read it

- The ΠY(t−1) term carries the equilibrium
- Each column of β is a cointegrating vector
- α is the adjustment speed of each series (Part 3)
- Engle-Granger finds at most one relation

Rank = number of cointegrating relations

The trace test sets the rank step by step

Start at r = 0 and stop at the first null you cannot reject

What rank r means (k series)

$$r = 0: \text{ no relation} \qquad r = k: \text{ all } I(0)$$
$$0 < r < k: \; r \text{ cointegrating relations}$$

Trace and max-eigenvalue statistics

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{k} \ln\left(1 - \hat{\lambda}_i\right)$$
$$\lambda_{\max}(r) = -T \ln\left(1 - \hat{\lambda}_{r+1}\right)$$

How to read it

- λ̂: eigenvalues from Π (0–1); larger means a stronger stationary direction
- Trace: at most r vs more than r
- Max-eigenvalue: exactly r vs r + 1
- With k series, at most k − 1 relations

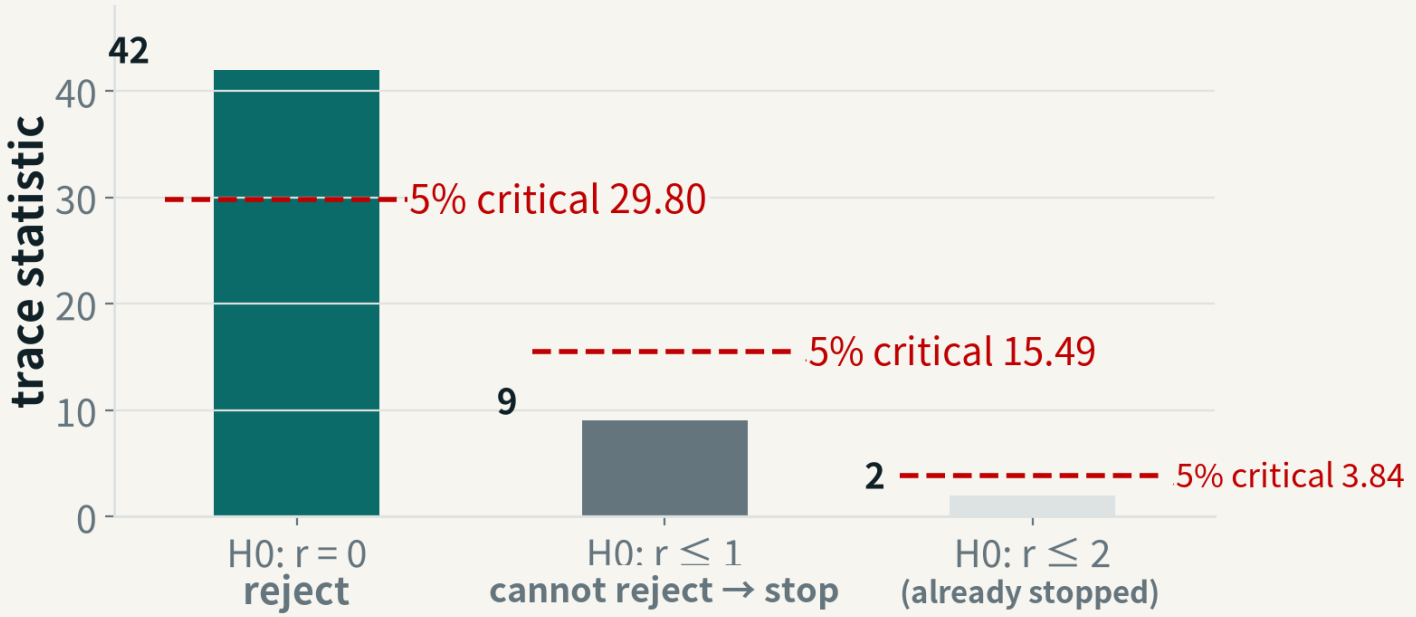
If the two disagree, dig into why

Only r = 0 is rejected: the rank is 1

Trace statistics for three series (about 42, 9, 2 from the course figure) and standard 5% critical values

READING THE TRACE TEST

- 42 > 29.80 → reject r = 0
- 9 < 15.49 → cannot reject r ≤ 1
- Stop here: rank 1
- One independent long-run equilibrium



For a single pair, the Johansen rank must be 0 or 1

A pair: Engle-Granger. A system: Johansen

Both ask the same question — 'is there a stationary combination?' — in different ways

Feature	Engle-Granger	Johansen
Series	2 (a pair)	2 or more
Approach	Single equation, two steps	System (VECM-based)
Relations found	At most 1	At most k − 1 (the rank)
Symmetric?	No — pick a dependent	Yes

※ Traps: structural breaks, short samples (low power), lag choice — check robustness across lags, orderings and subsamples

A positive result gives β and the spread — the raw material for error correction and trading

03

Error Correction and Pairs Trading

Measure the force that pulls the spread back, then trade it

The ECM explains the next change by last period's gap

Two-variable ECM — a gap above 0 is pulled down, below 0 pulled up

Two-variable error-correction model (ECM)

$$\Delta Y_t = \alpha_Y \underbrace{(Y_{t-1} - \beta X_{t-1})}_{z_{t-1}\ (\text{last gap})} + \gamma_Y \Delta X_{t-1} + \varepsilon_{Y,t}$$

$$\Delta X_t = \alpha_X (Y_{t-1} - \beta X_{t-1}) + \gamma_X \Delta Y_{t-1} + \varepsilon_{X,t}$$

How to read it

- $z(t-1) = Y - \beta X$: last period's equilibrium error
- α: how fast each series reacts to the gap
- γ terms: short-run effects, separate from the long run
- Every term is stationary, so no spurious regression

No error-correction term, no cointegration

A VECM is a differenced VAR plus one correction term

It puts back, as an explicit correction term, the long run that differencing drops

VECM

$$\Delta \mathbf{Y}_t = \alpha \underbrace{\left(\beta^\top \mathbf{Y}_{t-1}\right)}_{z_{t-1}} + \sum_{i=1}^{p-1} \Gamma_i \, \Delta \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t$$

The spread's own adjustment speed

$$\Delta z_t = \Delta Y_t - \beta \, \Delta X_t = \underbrace{(\alpha_Y - \beta \, \alpha_X)}_{\alpha_z < 0} \, z_{t-1} + \cdots$$

How to read it

- Γ terms: short-run momentum and noise (an ordinary VAR)
- α(β′Y) term: all of the long-run behaviour
- $\alpha < 0$ is needed for the gap to close
- The series with larger $|\alpha|$ does most of the correcting

The only difference from Week 6's VAR is this correction term

The half-life turns adjustment speed into days

Ignoring short-run terms, the spread is AR(1) — a share ρ of the gap remains each period

The spread as an AR(1)

$$z_t = (1+\alpha_z)\, z_{t-1} + u_t = \rho\, z_{t-1} + u_t$$
$$-1 < \alpha_z < 0 \iff 0 < \rho < 1$$

Half-life

$$\rho^h = \tfrac{1}{2} \implies h_{1/2} = \frac{\ln 0.5}{\ln(1+\alpha_z)} = \frac{\ln 0.5}{\ln \rho}$$

How to read it

- ρ = 0.90 (α = −0.10) gives a half-life of about 6.6 days
- Sets a sensible holding period
- Screens out spreads that are too slow
- Checks the estimated speed is plausible

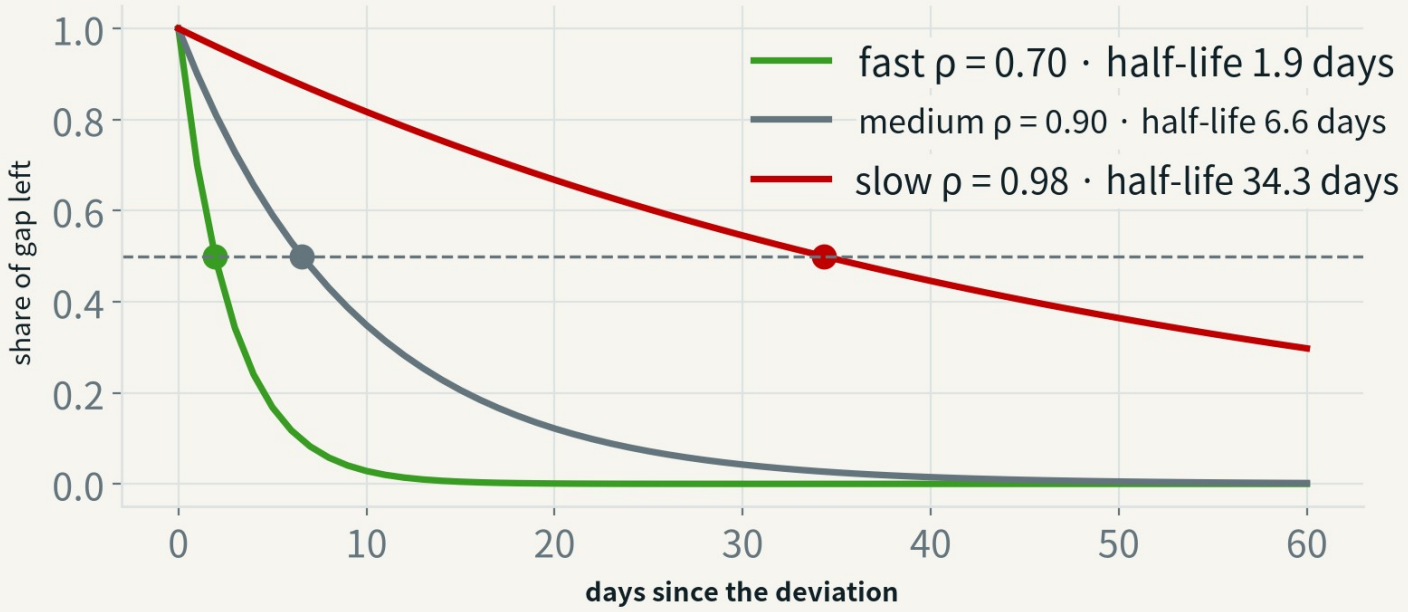
Between 3-day and 60-day half-lives, the 3-day one is more tradable

Half the gap closes in 2 days at ρ 0.70, 34 at 0.98

Share of the gap left after a deviation; dots mark each half-life

FAST vs SLOW

- Fast ρ 0.70: 1.9 days
- Medium ρ 0.90: 6.6 days
- Slow ρ 0.98: 34.3 days
- The slower, the longer capital is tied up



Fast, reliable reversion is what makes a trade worth doing

Cointegration and ECM come together; signal by z-score

Granger representation theorem → entry and exit rules for pairs trading

Granger representation theorem

$$\text{cointegrated} \iff \text{error-correction representation exists}$$

z-score signal

$$Z_t = \frac{z_t - \mu_z}{\sigma_z}, \qquad |Z_t| \geq 2: \text{ enter}, \qquad Z_t \to 0: \text{ exit}$$

How to read it

- Never one without the other — why we use a VECM
- $Z \geq +2$: sell the spread (sell Y, buy βX)
- $Z \leq -2$: buy the spread; exit when Z returns to 0
- Estimate μ and σ from a past window only (no look-ahead)

Bet against the stretch, collect when it snaps back

Enter at ±2σ, exit at 0: P&L builds in steps

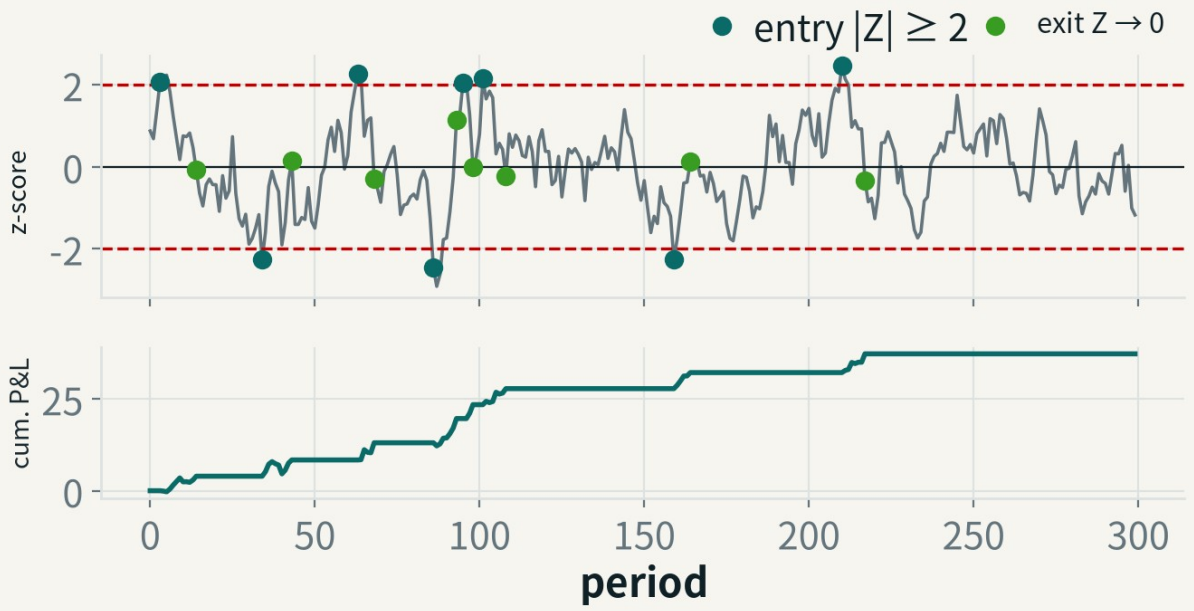
Simulated spread with ρ = 0.9, 300 periods, 8 entries, P&L in spread units (illustrative data)

Position and P&L

$$\text{position} = \pm\,(1 \text{ unit of } Y,\ -\beta \text{ units of } X)$$
$$\text{P\&L}_t = \text{pos}_{t-1}\,\Delta z_t, \qquad \text{size} \propto \sigma_z^{-1}$$

PAIRS TRADING STAGE II

- Market-neutral with the β ratio
- Size by spread volatility



The bet is on the spread, not the market

When the relation breaks, the spread drifts away

Mean-reverting AR(1) at first, a trending random walk after t = 180 (illustrative data)

WHEN IT BREAKS

- Causes: mergers, regulation, business-model shifts
- The old β no longer fits
- Defences: stop-loss, regular re-testing, many pairs
- Cointegration is a working assumption, not a law



Pairs trading without a stop-loss is a bet on a random walk

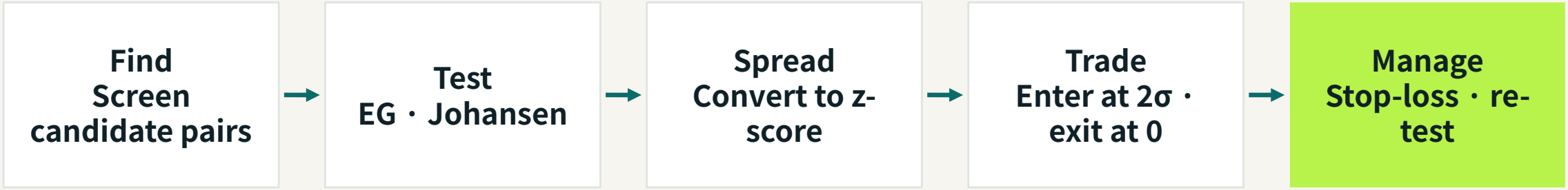
04

Synthesis

Concept → test → trade, and the ever-present risk of breakdown

Three steps: find, prove, trade

Pairs-trading pipeline — engine below: cointegration (anchor), error correction (pull back), volatility (bands and size)



Part	Key idea
Part 1 Cointegr.	One common trend → stationary spread = long-run equilibrium
Part 2 Tests	EG residual test · Johansen rank
Part 3 Trading	α · half-life · z-score · breakdown risk

Next, Week 8: state-space models and the Kalman filter

Find the anchor — cointegration

Prove the anchor — EG · Johansen

Trade the pull back — error correction

Next week
State-space models and the Kalman filter